Tasmin Khan

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Education

Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering (EEE)
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

- CGPA: **3.80 / 4.00** Feb'2020 - Mar'2025
- **Dean's List Award**, BUET (Awarded for Average GPA $\geq$ 3.75 in two consecutive terms) 2023, 2021
- **University Merit Scholarship**, BUET (Awarded based on merit ranking in term examinations) 2021

Relevant Coursework

- VLSI Circuits and Design [Lab]
- Analog Integrated Circuits & Design [Lab]
- Digital Electronics [Lab]
- Microprocessor & Embedded Systems [Lab]

Publications

• Priority Aware Deficit Round-Robin Arbitration for Fair Per Bandwidth AXI4 Crossbar Sept'2026
Submitted to Int'l Conf. on Electrical and Computer Engineering (ICECE)
Tasmin Khan, Ch. Rakin Haider, Tishna Sabrina

• A CNN Neuron for Always-On Near-Sensor Inference in 7-nm CNFET Aug'2026
Submitted to Int'l Conf. on Electrical and Computer Engineering (ICECE)
Tasmin Khan, Tanvir Hossain, Md Hasib Ur Rashid, Ahmed Zubair

• Design and Development of Low-Cost Earthquake Monitoring System Based on MEMS Sept'2026
IEEE Transactions on Instrumentation and Measurement DOI: 10.1109/TIM.2026.3743993
Tasmin Khan, Rahat Ibn Nabi, Nilotpaul Kundu Dhrubo et al

• Development of Low-Cost Seismic Monitoring System for Field Application in Bangladesh Aug - Sept'2025
Presented at IAGA/IASPEI Joint Scientific Meeting | Lisbon, Portugal
Al-Hussaini, T.M., Baten, M.Z., Nabi, R.I., Islam, K., Ghosh, P., **Khan, T.** et al

Research Experience

Hardware Architecture & RTL Automation **Ongoing**
Principal Investigator: Dr. Rakin Haider, Assistant professor, Computer Science and Engineering (CSE), BUET
Funded by: Bangladesh Industry Research, Development and Innovation (BIRDI)

- Leading design development: authoring design specifications, delegating tasks, and designing RTL with testbenches including AMBA bus interconnect, memory controller and systolic array for GEMM operation
- Exploring sparsity-enabling architectures for inference hardware
- Led training sessions teaching the research team SystemVerilog, hardware design, and testbench methodology
- Investigating limitations of existing datasets (VerilogDB, VeriCoder, RealBench) for industrial-scale designs
- Testing staged spec-to-RTL generation via intermediate representations to improve LLM design performance

Development of Low-Cost Methods for Monitoring and Early Warning for Geo-hazards [GitHub, Live]
Principal Investigator: Dr. Tahmeed Malik Al-Hussaini, Professor, Civil Engineering (CE), BUET
Co-Principal Investigator: Dr. Md Zunaid Baten, Associate Professor, EEE, BUET
Funded by: Research and Innovation Center for Science and Engineering (RISE) Internal Research Grant, BUET

- Conducted noise analysis of MEMS sensors, determining the optimal device for signals down to 0.1 m/s²
- Built stand-alone unit, validated in lab, and field-deployed a multi-station network detecting multiple earthquakes

Search Towards an Advanced Deep Learning Algorithm for Magnetic Resonance Imaging (MRI) Quantitative Susceptibility Mapping (QSM) with Magnetic Anisotropy (UG Thesis) [GitHub]
Supervisor: Dr. Maruf Ahmed, Assistant Professor, EEE, BUET

- Developed 2D & 3D datasets by implementing a complete multi-orientation MRI preprocessing pipeline
- Implemented CNN and Transformer models with PyTorch, finding CNNs more suitable due to localized anisotropy

VLSI Competition

Champion in the VLSI-thon arranged by Neural Semiconductor Dec'2024

- Identified the error source in a circuit from the output clock pattern and designed the solution logic circuit
- Designed the circuit and layout to implement logic from a given truth table, along with DRC and LVS checks

Technical Skills

Design & Implementation: Cadence (Virtuoso, Genus, Innovus, Assura), Vivado, Quartus, OrCAD PSpice, Proteus
Simulation & Verification: Xcelium, ModelSim, QuestaSim, Icarus Verilog, Aldec Riviera-PRO, Cocotb, Pytest
Programming & Modeling: Verilog HDL, SystemVerilog, Python, C, C++, PyTorch, Git, MATLAB, Simulink
Development Platforms: Linux, Keil uVision, Makefile, Arduino IDE, FPGA and CPLD Development Boards

Industrial Experience

Modeling Engineer **Apr'2026 - Present**
gixsystems-AI, Dhaka, Bangladesh

- Behavioral Modeling and Design of Power Converter Integrated circuit and Discrete Devices
- Built a physics-informed DL pipeline mapping characteristic curves to compact model parameters

Internship on VLSI Design Jun - Jul'2024
Silliconova, Dhaka, Bangladesh

- Study of Digital Design and Verification, Physical Design, DFT, Packaging and FinFET, GAA, CFET technologies
- Hands-on experience in standard cell Layout Design, DRC/LVS Verification, and Placement Techniques

Projects

• **Three Band Digital Audio Equalizer Peripheral with biquad IIR filters** [GitHub]
Designed three-band audio equalizer with 16-bit fixed-point biquad IIR filters compiled using Icarus Verilog and verified with cocotb Python testbenches, implementing a RISC-V peripheral for digital audio processing

• **16bit Ripple Carry Adder with Coverage Driven Layered Testbench and Layout** [GitHub]
Developed a layered testbench in NCSim/Xcelium skewing randomization toward unhit coverbins, and performed PnR in Innovus, achieving faster coverage closure and a 93.71% density-utilized layout

• **Hardware Gaming Interface to play Single Round Tic-Tac-Toe with Custom PCB** [GitHub]
Built game logic with player identification and cheat prevention designing PCBs in Proteus for an error-free play

Teaching Experience

Adjunct Lecturer Dec'2025 - Apr'2026
Department of Electrical and Electronic Engineering (EEE)
Ahsanullah University Of Science and Technology (AUST), Dhaka, Bangladesh

Physics & ICT Instructor 2021-2023
Udvash Academic Coaching, Grade: 8 - 12

Awards

- **RISE Undergraduate Student Research Grant** of ∼700 USD for Thesis 2024
- **Honorable mention** in Intra BUET PCB Design Contest arranged by IEEE RAS, BUET 2023
- **Second runner-up** in 'HULT Prize at BUET' - A Social Entrepreneurship Competition 2020
- **Government Merit Scholarship**, Ministry of Education, Bangladesh 2019, 2017

Standardized Test

IELTS Academic: Overall 8.5 | Listening: 9.0 | Reading: 9.0 | Writing: 8.0 | Speaking: 8.0 Aug'2025

Leadership Activities

- Chairperson, IEEE WIE Affinity Group, BUET Student Branch 2024
- Assistant General Secretary, IEEE WIE Affinity Group, BUET Student Branch 2024
- Executive Member, IEEE Bangladesh Section, BUET Student Branch 2021-22